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FM: Flexible mapping from one gesture to multiple semantics

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ABSTRACT

An algorithm for flexible mapping (FM) from one gesture to multiple semantics in the same situational context is presented for the first time to reduce the cognitive and operating loads of an operator by using gesture commands. First, the foundation of the FM is built with the semantic-oriented difference features of behavior model (SDFBM) of the operator as implicit inputs. Second, the gesture–semantics FM algorithm and its extended process, namely, attribute classification for FM are proposed and implemented. Third, five commonly used gestures are designed to demonstrate the way a gesture is mapped to multiple semantics. Finally, several comparative experimental results are provided to demonstrate the superiority of the proposed methods to the state of the art. The main innovation of this study is that the same gesture in the same context can be mapped to several different semantics through the SDFBM feature recognition. This study provides an intelligent and natural interaction interface model for 3D platforms and key support for gesture-based implicit interaction design. The proposed algorithms are also tested or used in several applications, such as smart teaching interface and onboard vehicular systems and intelligent TV.

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1. Introduction

In this modern age, advancements in ubiquitous computing have required the use of natural user interface. The presence of computers and use of the facilities of human–computer interaction in our societies will positively influence our societies.

Users can control an interaction with hand gestures in two ways, that is, based on MEMS (microelectromechanical systems) inertial sensor and digital image processing technologies. A MEMS inertial sensor technology-based gesture controller represents a "mouse in the air." In image processing-based interaction, gestures are captured with image sensors; then, image analysis technology is used to track and identify the hand motion of a user; this gesture is further converted to related instructions and operations through a computer. Free gestures are more natural than the MEMS when pointing, hitting, reaching, or throwing is required. However, comparisons of gestures across the two scenarios have not been performed despite the simultaneous existence of the two approaches for several decades and the recent increased popularity for use in video games (e.g., the Nintendo Wii, Sony Move, and Microsoft Kinect), [36]. Human–computer interaction with hand gestures plays a significant role in these modalities because humans frequently rely on their hands in communication or

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interaction with their environments. Therefore, hand gesture-based methods become prominent over other approaches by providing a natural means of interacting and communicating. However, most users complain about currently widespread issues in gesture-based interfaces, such as difficulties in error correction and memorizing gesture commands.

A gesture-based onboard vehicular system enables a driver to operate the control panel for navigation maps or entertainment systems. However, this activity may place heavy extra memory and operational loads on drivers because drivers may be required to remember several gestures. Any extra loads for drivers should be cautious because these drivers are required to focus on traffic. These same issues remain in gesture-based smart classroom interface (SCI) system. The SCI system is composed of a computer, large screen, projector, Kinect, and gesture interaction algorithms. Teachers can select, scale, rotate, and translate selected objects on the screens of the SCI system. The teachers should concentrate their energies on teaching contents but not recall different gestures or interactions in teaching activities. The third field with the same issues is gesture-based intelligent TV (iTV) system. Controllers of iTV are mostly elderlies, and requiring them to remember numerous gestures and operating them for TV control is inconsistent with human-centered design ideas because it places heavy user loads on the elderlies. Numerous examples can be provided, but we omitted them given the limited length of this paper. Owing to such type of issues, this study focuses on building a flexible mapping (FM) algorithm between gestures and semantics to reduce the cognitive and operating loads of the operator.

2. Related work

Gestures can be classified as static [8] and dynamic [32]. Static hand gestures (hand postures/hand poses) are motions in which the hand position remain unchanged during the gesturing period, and these gestures mainly rely on the shape and flexure angles of the fingers. In dynamic hand gestures, the hand position changes continuously with respect to time. Dynamic gestures rely on hand trajectories and orientations, in addition to the shape and flex angles of the fingers.

Human–computer interaction has regained popularity recently given the intuitive and successful interaction techniques of devices, such as tablet PCs, smartphones, and even smart houses. Vatavu presented the results of the first study that addresses the free-hand gesture preferences of users for TV control and suggested 12 common TV control tasks in designing free-hand gesture-based interfaces for the next generation of home entertainment systems and smart living room environments [35]. Vatavu and Zaiti [38] collected 378 gestures with corresponding fit-to-function ratings and assigned each referent with the gesture that received the highest agreement. In the earlier studies, only a few gestures were suggested based on the specialized knowledge of the designers [2,27], and several of these gestures are difficult to discover and adopt because they are arbitrarily associated with commands [43]. Afterward, the number of gesture-activated motivations has increased, and intuitiveness has become one of the most important considerations in gesture-based interaction [29]. Therefore, the issue has expanded to determining the gestures for further intuitive interaction with products, and various studies have started to focus on matching gestures with commands for user-centered interaction [17,22].

The second common problem is gesture recognition with high recognition ratio. The recognition of gestures is a comprehensive task that combines various aspects of computer science, such as motion modeling, motion analysis, pattern recognition, and machine learning. Most of the static gesture recognition research focuses on neural network-centered approaches [26,28]. Neßelrath et al. [24] used the WiiMote remote control, which can detect direction, position, and motion, as the sensing device and a multilayer perceptron neural network, support vector machine (SVM), and dynamic time warping algorithm to achieve gesture detection and recognition, thereby enabling the interaction among seven types of smart home equipment. Kim et al. [16] used the Kinect as an input device to provide the same touch interface as smartphones at long distances. Lee et al. [19] combined the Kanade–Lucas–Tomasi tracking algorithm and SVM technology to detect and identify gestures and found that the detection accuracy and recognition rate for five types of gestures that were performed 2 m away were 98% and 80%, respectively. Zhao et al. [44] proposed a recursive induction learning based on extended variable-valued logic for hand pose recognition; the proposed algorithm can automatically select the most effective features, thereby confirming the suitability of the proposed algorithm for real-time gesture recognition systems. Huang et al. [9] proposed an algorithm for hand posture recognition using an SVM classifier under varying illumination and pose conditions, in which insensitivity to hand pose variations is gained with a Gabor filter-based pose angle estimation and correction method. Zhou et al. [45] presented a new finger detection algorithm and extracted salient hand edges for detecting extensional and flexional fingers through the parallel edge characteristics; in addition, angular projection centered on wrist position was used to obtain the angle, orientation, and length of each finger. This method can directly extract a high-level hand feature and estimate hand poses in real-time.

For dynamic gesture recognition, hidden Markov model (HMM)-based approaches are generally preferred because these approaches yield improved results [6,23,43]. Dias et al. [5] required users to use five types of gestures (wave, swipe up, swipe right, push, and pull) to interact with a smart TV. Bartolini et al. [1] used seven types of gestures (up, right, down, left, cir, sq, and x) to control smart environments; these gestures begin and end with the user holding the cube in the same position and are executed on the vertical plane in front of the user, who holds the device every time with the same orientation. Yeasin and Chaudhri [42] utilized finite-state machine-based techniques to recognize dynamic gestures. Different from these methods, Kılıboz and Ug˘ur Güdükbay [14] presented an intuitive approach to teaching a machine to recognize a hand gesture command for users to apply this command to devices such as TVs or eReaders. Motion data in gradient-like form were collected through this method and allowed users to create their own gesture commands for a particular task in accordance with the perception of these users on gestures that suit the action. Chen et al. [3] utilized Fourier descriptor

and optical flow-based motion analysis to characterize spatial and temporal features, respectively. The algorithm extracts the shape of a hand from complex backgrounds by tracking the hand in real time. HMM-based recognizers identify the optimal likelihood gesture model for a given pattern. Lee and Kim [18] introduced a HMM-based threshold model concept to filter patterns with minimal likelihood. Marcel et al. [21] proposed an extension of the HMM, namely, input/output HMM (IOHMM), for HGR. The IOHMM is based on a non-homogeneous Markov chain in which emission and transition probabilities depend on the input. Shin et al. [34] utilized Bezier curves for trajectory analysis and classification of dynamic gestures. Gestures are recognized by fitting the curve to the 3D motion trajectory of the hand. The gesture speed is incorporated into the algorithm to enable accurate recognition from trajectories with varying speed.

Another issue relates to the user workloads. Nguyen et al. [25] established a gesture vocabulary database and allowed users to use five types of gestures, thereby requiring the users to remember 12 types of gesture commands to interact with a robot; this strategy resulted in a heavy memory load on the users. Rempel et al. [31] mentioned that users constantly suffer from hand pain during gesturing. These authors studied 24 professional sign language interpreters who reported discomfort during hand gestures that were associated with 47 characters and words and 33 hand postures. The widely used hand gestures are limited to a carefully selected vocabulary of symbolic gestures that are mostly used for issuing commands [30]. These methods require users to remember the vocabularies and the responding gestures, thus leading to heavy loads on the users. Schlömer et al. [33] reported on the performance of gesture recognition with a Wii controller. These authors tested their system using five 3D gestures and proposed that users should be allowed to select and train their own gestures. Jacob [11] suggested the calculation of the duration of the cursor on the interface objects by using a gesture control interface to determine whether the object was selected. This method clearly exhibits several deficiencies. The user must concentrate on the time of stay on the object when in operation; this activity increased the cognitive and operational burden for the user. Kato et al. [13] presented a strategy based on spatial proximity principle, which main idea was to complete a series of interactive tasks by determining the distance between the handheld paddle and the virtual objects. However, the distance between the paddle and the virtual object is difficult to control, hence causing false selections. Kjeldsen et al. [15] proposed establishing certain mapping relations between gestures and visual interface widget. This method may slightly solve the problem of false elections while the system must generate numerous widgets; however, this method will undoubtedly increase the cognitive load of users.

The third issue is the mapping between gestures and commands. Ad hoc maps between a predefined set of gestures and commands are widely used and adopted. The implementation of ad hoc recognizers can be challenging if the number of gestures is large because gestures “collide” in feature space. Ad hoc recognition also prevents application end-users from defining their own gestures at runtime [39]. These authors further extended this work by reporting user-dependent and user-independent recognition results [37]. Currently, most methods define the maps between gestures and commands based on frequency ratio. The users are instructed to derive a gesture for each command. Similar gestures for each command are gathered based on the physical shapes/motions once the gestures are collected from the users. Thus, the gestures with high frequency for each command are selected and estimated in terms of subjective measures, such as suitability, ease of memory, and fatigue. Finally, one of these motions is selected as the final gesture for the command. However, this method is likely to disregard the meaningful gestures given their low frequency. Choi et al. [4] formulated and verified the two hypotheses: (a) users may change their selection after observing other gestures and (b) a gesture derived from only a few users might be a favorable gesture. Their experimental results also implied that the frequency cannot guarantee the selected gestures because users in the first step have only a limited set of gestures in their mind.

Overall, the previously mentioned state of the art focuses on building the most commonly used and natural gesture set and mapping a gesture to a fixed command. Therefore, two basic problems are highlighted. First, users are not free to select the gesture. Second, users are not free to select the semantic meaning. In the current and past engineering approaches, we observed that the gesture meaning can be a position on a screen, an ordered list of reference objects computed from a pointing ray, an abstract model of a shape, and a style of movement. In most cases, the issue of semantic meaning is not addressed because the systems stop at the recognition stage.

In contrast to the state of the art, this study is aimed at probing the FM between gestures and semantics and assumes that the mapping between a gesture and a command is fixed. In the same situational context, a one-to-multiple mapping occurs between a gesture and various semantics, and a multiple-to-one mapping occurs between different gestures and the same semantic.

We define several basic concepts to describe the FM between gestures and semantics.

2.1. FM

For convenience of description, the FM between the same gestures to different semantics in the same context is called FM. For FM, one gesture is mapped to K ($K \geq 2$) semantics. FM is described with semantic-oriented difference features of behavior models (SDFBM).

2.2. SDFBM

Difference features are frequently observed among the behavior models of the gestures of an operator that can distinguish different semantics when using a single gesture to express different gesture semantics (here, these semantic sets that

can be expressed using the same gesture are called semantic groups (SG); each semantic group is assigned a unique semantic group number). In this study, these difference features are called SDFBM. For example, the gesture $\boldsymbol{g}$ = "fetching hand" can express the semantic set $\boldsymbol{ss}$ = {"fetch an object," "shrink an object"}. For the same gesture $\boldsymbol{g}$, if the semantics "fetch an object" and "shrink an object" can be distinguished by the moving speed v of the gesture $\boldsymbol{g}$, then $\boldsymbol{SDFBM_{ss}} = \{v\}$.

3. Base of FM

3.1. Basic cognitive experiments

We invited 50 college students aged 21–23 years to participate in the experiment to explore the rules of FM from one gesture to multiple semantics in the same context. The test subjects were required to have basic experience in operating smart TVs or smartphones. The test subjects wore a hand position tracking device and 5DT digital gloves, from which a computer could receive the data on gesture joint motions and angles through the interface.

The current operation object is assumed to be a cube, to which the operations of "fetch" and "shrink" can be applied. Based on the interaction experience that the operators obtained from operating the smartphones, both operations can be achieved by performing the gesture of "folding the fingers of an open hand into a fist" ($\boldsymbol{G_1}$), and all participants participated in the "fetch" and "shrink" experiments. The 3D scene displayed a 3D cube, and the operators performed 10 sets of "fetch the current object" operations. Participants were asked to use the most natural means of expressing different semantics sg = {"fetch the current object," "shrink the current object"} to obtain the mean changing speed of various gestures. For each person, each gesture of "fetch" and "shrink" was first repeated 10 times, and then the average speed of all the 15 joint angles of the gesture was computed, thereby generating a speed sample. For each gesture, we obtained a speed sample for each subject. Thus, 50 subjects produced 50 speed samples for each gesture. A total of 40 samples expressed the semantic of "fetch" out of the 50 speed samples for "fetch" gesture, while 20 samples expressed the semantic of "shrink" out of the 50 speed samples of "shrink" gesture.

3.2. Statistical data analysis

The experimental results of the task showed that 40 of the 50 speed samples, which represent 80% of all samples, expressed the semantic of "fetch" and were within the speed range of [2, 4]. A total of 20 of the 50 speed samples, which comprise 40% of all samples, expressed the semantic of "shrink" and were within the speed range of [1, 2]. Thus, the two semantics for gesture $\boldsymbol{G_1}$ in the same situational context can be distinguished statistically by comparing the speed of the gesture joint motion of a user: if the speed was in the range of [2, 4], then the gesture semantic was "fetch the current object," and if the speed was in the range of [1, 2], then the semantic gesture was "shrink the current object." Table I summarizes the experimental results for using the "fetching hand" gesture in expressing different semantics in the same situational context (additional experiments are presented in the appendix at the end of the paper). Thus, $\boldsymbol{SDFBM_{SG}} = \{speed\}$, (s_1, c_1) = ("fetch," [2, 4]) and (s_2, c_2) = ("shrink," [1, 2]). Here, the SG $\boldsymbol{S} = (s_1, s_2)$ = ("fetch," "shrink"), and the conditions of semantics $\mathrm{C} = (c_1, c_2)$ = ([2, 4], [1, 2]).

We also noticed that several SDFBMs overlap between s_1 and s_2. The number of samples with the speed 2 is 2 in the group of samples for s_1, and the number of samples with the speed 2 is 4 in the group of samples for s_2. Owing to the two groups of samples together, 6 of the 100 speed samples from s_1 and s_2 have features that overlap, thereby indicating that the overlap rate is 6% for the gesture $\boldsymbol{G_1}$. In a conversation with the subjects with speed 2, 4 of the 6 subjects stated that this situation constantly occurs if their interaction goal or semantics is ambiguous in their mind in the gesture operation. Furthermore, statistical analysis of the 50 samples demonstrates that s_1 obeys Gaussian distribution with an average speed of 2.96 and mean square error (MSE) of 0.8572 or $s_1 \sim \mathrm{N}(2.96, 0.8572^2)$. We obtain $s_2 \sim \mathrm{N}(2.08, 1.425^2)$ through the same method.

According to the results presented in the above experiments, the computer can distinguish among different semantics of the same gesture with the speed and trajectory as implicit inputs, and the SDFBM features can be used to realize the mapping between one gesture and multiple semantics. The FM between one gesture and multiple semantics can be established given the SDFBM features of the behavioral models of the operator as implicit inputs, that is, the foundation of the FM algorithm in the present study. Thus, the method that is used to effectively design the SG in the form of ***Rule R*** can be derived as follows:

R: The expression of the different semantics with the same gesture is possible given the same situational context of the interaction using SDFBM features, which can be constructed as the following cues: the trajectory scope, magnitude, or speed of gesture changes, will be significantly different if the same gestures express different semantics. In a special case, the semantic or gesture group can also be constructed based on the life experience or metaphor technology of the operator.

4. Gesture–Semantics FM algorithm

The following two basic datasets are requested. (1) The operator gesture database describes the basic gestures of the operator and is composed of gesture numbers and features for recognition. (2) The SDFBM feature database (SFD) describes the mapping between one gesture and multiple semantics. The SFD is composed of the following fields: object number,

Algorithm 1 FM algorithm.

While (interactive tasks are not accomplished) Do
{
1. The operator selects the operation object $\boldsymbol{O}$ in the scene.
2. Acquire the user's motor gesture $\boldsymbol{g}$ with gesture recognition algorithm from OGD dataset.
3. Retrieve the $SG_{\boldsymbol{O},\boldsymbol{g}}$ with entrance $\boldsymbol{O}$ and $\boldsymbol{g}$ from SFD dataset.
4. Retrieve the SDFBM $\boldsymbol{\varphi}$ and their conditions set $\mathbf{C_0}$ from SFD dataset.
5. If $c_i \in \boldsymbol{C}$ is satisfied, the $s_i \in \boldsymbol{S}$ is identified.
6. The gesture $\boldsymbol{g}$ interacts with the object $\boldsymbol{O}$ using semantic s_i.
}

Algorithm 2 Acquisition of gesture trajectory.

1. The dynamical model of the pose motion is determined as follows:

$$\boldsymbol{X}_k = \boldsymbol{f}(k-1, \boldsymbol{X}_{k-1}) + \boldsymbol{\eta}_{k-1}, \quad (1)$$

$$\boldsymbol{Z}_k = \boldsymbol{h}(k, \boldsymbol{X}_k) + \boldsymbol{V}_k. \quad (2)$$

2. $\boldsymbol{X_k}$ is estimated by the following:

$$\hat{\boldsymbol{X}}_k = \boldsymbol{f}(k-1, \boldsymbol{X}_{k-1}), \quad (3)$$

$$\boldsymbol{X}_k^{(i)} = \hat{\boldsymbol{X}}_k + \boldsymbol{\eta}_k, \quad (4)$$

$$\boldsymbol{X_k} = \sum_{i=1}^{N_k} \omega_k^{(i)} \boldsymbol{X_k^{(i)}}, \quad (5)$$

$$A_k^i = e^{-\lambda H_m}, \quad (6)$$

$$\omega_k^{(i)} = \frac{A_k^i}{\sum_{s=1}^{N_k} A_k^s}, \quad (7)$$

$$\omega_t = e^{-\lambda H_{(t)}}, \quad (8)$$

3. $\boldsymbol{X_k}$ is rendered and displayed.

gesture number, SDFBM set $\boldsymbol{\varphi}$, SG $\boldsymbol{S}$, and condition set $\boldsymbol{C}$ for each gesture, semantic, and corresponding condition (s_i, c_i), among which $s_i \in \boldsymbol{S}$ and $c_i \in \boldsymbol{C}$. First, the interaction context is sensed with the selected object and the gesture of the user, and we place the flexible mapping in the same context from one gesture to multiple semantics into the SFD dataset. Hence, SG can be acquired by observing the SFD dataset. Next, the SDFBM features of the gesture of the user are extracted from the gesture image sequences, and the conditions that the features are satisfied are computed; thus, the corresponding semantics are retrieved from the SFD dataset. Lastly, the semantics are exerted on the selected object. Based on this idea, the proposed FM algorithm is described as follows.

Algorithm 1.

5. Gesture recognition

Huang and Zhou [10] used density distribution feature (DDF) as gesture features and Euclidean distance as similarity operator to overcome the influence of rotation, translation, and scaling. However, differentiating two similar gestures is difficult through this method. Hence, Yang et al. [41] used hand coordinate distribution features as gesture features and Hausdorff-like distance as similarity operator. The abovementioned algorithms are mainly effective for static gesture recognition. Therefore, we propose a dynamic gesture recognition algorithm based on Hausdorff-like distance.

5.1. Gesture trajectory recognition

At time k, If $\boldsymbol{X_k}$ is the 3D position vector with the dimension of 3×1 of gesture trajectory, then the predicted value of $\boldsymbol{X_k}$ is assumed $\hat{\boldsymbol{X}}_k$, and $\boldsymbol{\eta}_k$ is a white noise random vector.

In Algorithm 2, $\boldsymbol{f}$ and $\boldsymbol{g}$ are motion and observation models of gesture trajectory, respectively; ω^i refers to the weight of a particle; N_k is the number of the particles at time k; λ is an empirical constant that is set as 0.01. H refers to the Hausdorff distance [9] between the observation values $\boldsymbol{Z_k}$ of the posture and projections of the 3D hand model into the frame image.

The obtained gesture trajectory is changed into a fitting curve using least squares method [12] and is recognized by using Hausdorff distance.

5.2. Dynamic gesture recognition

We first translate a dynamic sequence of a gesture into a static image and then recognize the static image based on Hausdorff-like distance. For the static image, we divide the hand pose into SRs, that is, the region between two adjacent circumcircles. We compute the Hausdorff-like distance of SRs between the query image from the input and the sample image from the gesture dataset after determining the center of the concentric circles as the center of gravity of the hand silhouette. For each pose, five frame images from different viewpoints are placed in the gesture dataset. Thus, the recognition is seldom affected by the viewpoint.

Algorithm 3 Dynamic gesture recognition algorithm.

1. The hand silhouette is segmented [20] and resized to 40×40 for each frame image of the query gesture $\boldsymbol{QG}$.
2. Combine the images of the query dynamic gesture into a static image as follows:
$$\boldsymbol{QG} = \cup \boldsymbol{HS} \tag{9}$$
Here, $\boldsymbol{HS}$ means a frame image of the query dynamic gesture, and $\cup\boldsymbol{HS}$ is a static image which is fused with some of the frame images of dynamic gesture sequence.
3. The image $\boldsymbol{QG}$ is resized to 40×40 and its centre of gravity $\boldsymbol{Z}$ is computed.
4. Figure out the maximum distance D_{max} between pixels of the hand silhouette and point $\boldsymbol{Z}$:
$$D_{\max} = \max\{\boldsymbol{Z}, \boldsymbol{QG}\}, \tag{10}$$
5. Figure out the maximum inscribed circle $\boldsymbol{C}$ with centre $\boldsymbol{Z}$ and radius D_{max}.
6. Subdivide the region bounded by the circumcircle $\boldsymbol{C}$ into K co-centric circular bands that are bounded by two co-centric circles with a radius of $(i-1) * (D_{max}/K)$ and $i * (D_{max}/K)$, the band is noted as $\boldsymbol{SR}$. $i = 1, 2, \ldots, K$.
7. Extract the pixels set for each $\boldsymbol{SR}$.
8. Compute the Hausdorff-like distance [37] $d(\boldsymbol{SR_{q,i}}, \boldsymbol{SR_{q,j}})$ between $\boldsymbol{SR_{q,i}}$ and $\boldsymbol{SR_{q,j}}$ as follows:
$$\tilde{d}(\boldsymbol{SR_{q,i}}, \boldsymbol{SR_{q,j}}) = \sqrt{\frac{1}{\|\boldsymbol{SR_{q,i}}\|} \sum_{(x,y)\in \boldsymbol{SR_{q,i}}} \min_{(u,v)\in \boldsymbol{SR_{q,j}}} ((u-x)^2 + (v-y)^2)}, \tag{11}$$
$$d(\boldsymbol{SR_{q,i}}, \boldsymbol{SR_{q,j}}) = \tilde{d}(\boldsymbol{SR_{q,i}}, \boldsymbol{SR_{q,j}}) + \tilde{d}(\boldsymbol{SR_{q,j}}, \boldsymbol{SR_{q,i}}), \tag{12}$$
here $q = 0, 1, \ldots, K$ and $\boldsymbol{SR_{q,i}}$ refers to the set composed of the skin pixels in the qth SR for the sample image.
9. Find the biggest Hausdorff-like distance HD_{max} out of the K $\boldsymbol{SR}$:
$$HD_{max} = \max(\mathrm{d}(\boldsymbol{SR_{1,i}}, \boldsymbol{SR_{1,j}}), \mathrm{d}(\boldsymbol{SR_{2,i}}, \boldsymbol{SR_{2,j}}), \ldots, \mathrm{d}(\boldsymbol{SR_{K,i}}, \boldsymbol{SR_{K,j}})) \tag{13}$$
10. Estimate the error $\boldsymbol{E}$ among the two images i and j using the formula (See Fig. 1):
$$\boldsymbol{E}_{pose}^{i,j} = \frac{1}{HD_{max}} \sum_{q=0}^{K} \omega_{\mathrm{q}} \boldsymbol{d}(\boldsymbol{SR}_{\mathrm{q,i}}, \boldsymbol{SR}_{\mathrm{q,j}}), \tag{14}$$
in which ω_q refers to weight of the qth SR. The greater the radius, the greater the weight.
11. Estimate the query gesture and the sample gesture out of the gesture dataset by obtaining the error $\boldsymbol{E}_{gesture}$ of a dynamic gesture using the formula:
$$\boldsymbol{E}_g = \sum_{w=1}^{\lfloor n/2 \rfloor} (\boldsymbol{E}_{pose}^{n-w+1} - \boldsymbol{E}_{pose}^{w})^2, \tag{15}$$
where n is the frame length of the dynamic gesture.
12. Return the gesture with the minimum $\boldsymbol{E}_g$ in the gesture dataset.

Algorithm 3 describes the detailed method of gesture dataset retrieval from the gesture dataset [7].

A typical method for realizing Formula (9) is combining the first, middle, and last frame images of a dynamic gesture sequence into an image in order. The two images, namely, $\boldsymbol{U}_{m\times n}$ and $\boldsymbol{V}_{m\times n}$, can be fused into an image with the following formula:

$$\boldsymbol{QG}_{i,j} = \begin{cases} \boldsymbol{U}_{i,j} & (0 \le j \le n-1) \\ \boldsymbol{V}_{i,j} & (n \le j \le 2n-1) \end{cases} \tag{16}$$

The gesture is defined as the combination of gesture trajectory and dynamic gesture. The gesture algorithm is composed of Algorithms 2 and 3, and the gesture is recognized using the two steps. The proposed algorithm is called fusion recognition (FR) because our method fuses recognition of trajectory and dynamic gesture.

5.3. Discussion of FM

The search for the SDFBM feature is the key to mapping flexibly from one gesture to multiple semantics in the same context. The following five types of SDFBM features are used in this study: folding speed of fingers presented in Table 1 in Section 3.2, global motion speed of gesture and trajectory diameter displayed in Table 4 in the Appendix and Example 3 in Section 6.2.2, and moving time and moving depth in Example 2 in Section 6.2.2. The selection of the SDFBM features depends on interactive situational context and experience of ordinary life. The SDFBM that is consistent with the mental model of the user should be prioritized in the selection. Furthermore, the features to be selected for the different semantics of the same gesture in the same context should be derived from the motion attributes of gestures in which moving speed, amplitude, and direction may be the most common considerations. Therefore, the behavioral model of the user for the same gesture in the same context should be first built and analyzed, and then motion attributes that are consistent with the mental model of the behavioral model are extracted. Lastly, the selected attribute is further classified, thereby allowing the different ranges of attributes to correspond to different semantics to adapt these features to application scenarios.

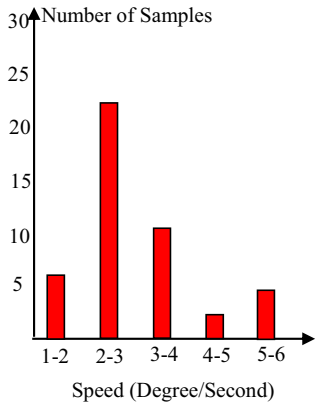
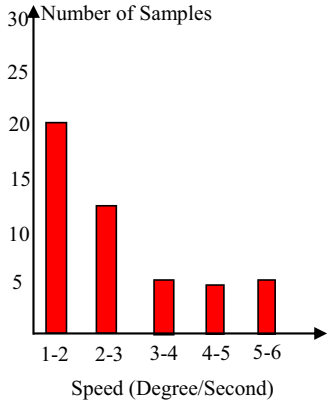
Based on the above discussion, we will further optimize the proposed FM in the following section.

5.4. Further improvement of FM

5.4.1. Problem description

In a specific context, for a gesture $\boldsymbol{g}$ with the attribute α and semantics set $\boldsymbol{S} = \{s_1, s_2, \ldots s_m\}$, we must find a mapping between α and $\boldsymbol{S}$ by satisfying the following conditions: (a) s_i and s_j ($i \neq j$) can be effectively separated from each other for the same gesture $\boldsymbol{g}$; (b) the mapping should be consistent with the mental model of the user with low cognitive load.

Table 1
Experiment for cognitive base of FM: SDFBM features of the gesture $\boldsymbol{G_1}$.

Operators' gestures	Semantics of the gestures $\boldsymbol{G_1}$	
	Fetch the current object (s_1)	Shrink the current object (s_2)
Fetching hand ($\boldsymbol{G_1}$)		

Algorithm 4 Attribute classification for FM (ACFM) algorithm.

1. Dynamic gesture recognition for gesture $\boldsymbol{g}$.
2. Define semantics set $\boldsymbol{S}=\{s_1, s_2, \ldots s_n\}$ of $\boldsymbol{g}$.
3. Sample data of some attribute α.
4. Make clustering analysis the data of the attribute α, acquiring n classes $\{c_1, c_2, \ldots c_n\}$ and each class is described with a normal distribution model $c_i \sim N(\mu_i, \delta_i^2)$. Here, $1 \le i \le m$ and $m \le n$.
5. Select m class $\{c_1, c_2, \ldots c_m\}$ form the n classes and compute the following conditions:
$$\begin{cases} \underset{i}{Max}\,|\delta_i| \le \delta_0 \\ |\mu_i - \mu_{i+1}| \ge \mu_0 \end{cases} \tag{17}$$
6. If the formula (17) is satisfied the mapping between $\{c_1, c_2, \ldots c_m\}$ and $\{s_1, s_2, \ldots s_m\}$ is found else the mapping does not exist.
7. Return.

5.4.2. Attribute classification for FM (ACFM)

We further propose the ACFM algorithm. The main idea of the ACFM is based on Gauss multimodal analysis of distribution for attribute α (Fig. 2), which is described by using Algorithm 4.

In the ACFM algorithm, we suppose that c_i obeys a normal distribution with mean μ_i and mean square error δ_i. δ_0 and μ_0 are empirical thresholds.

6. Experiment results and analysis

6.1. Settings

For the experimental settings, we used the Kinect, which captures 640 × 480 videos at 30 Hz. Our computer was equipped with Intel(R) Core(TM) 2 Quad-core CPU with a 2.66 GHz processor and 3.25GB of memory. A total of 50 junior college students who had smartphone operation experience were invited to join the experiments. For each subject, each semantic of a gesture was first repeated 10 times, and then the average SDFBM features were computed to generate a SDFBM sample. Kinect is commonly used in a free-hand gesture-based interaction to detect gestures; then, the different gestures are mapped to different commands to achieve the non-contact remote control of TVs [35]. This method essentially uses gesture commands to control interactions; thus, this method is called GC, which consists of four steps [40]. First, a posture image is segmented for each frame of a hand gesture from the camera video. Second, the gesture is recognized. Third, the command that corresponds to the gesture is retrieved. Finally, the command is interpreted and implemented. Therefore, GC is characterized by a one-to-one mapping between gestures and commands. The proposed FM algorithm will be compared with GC.

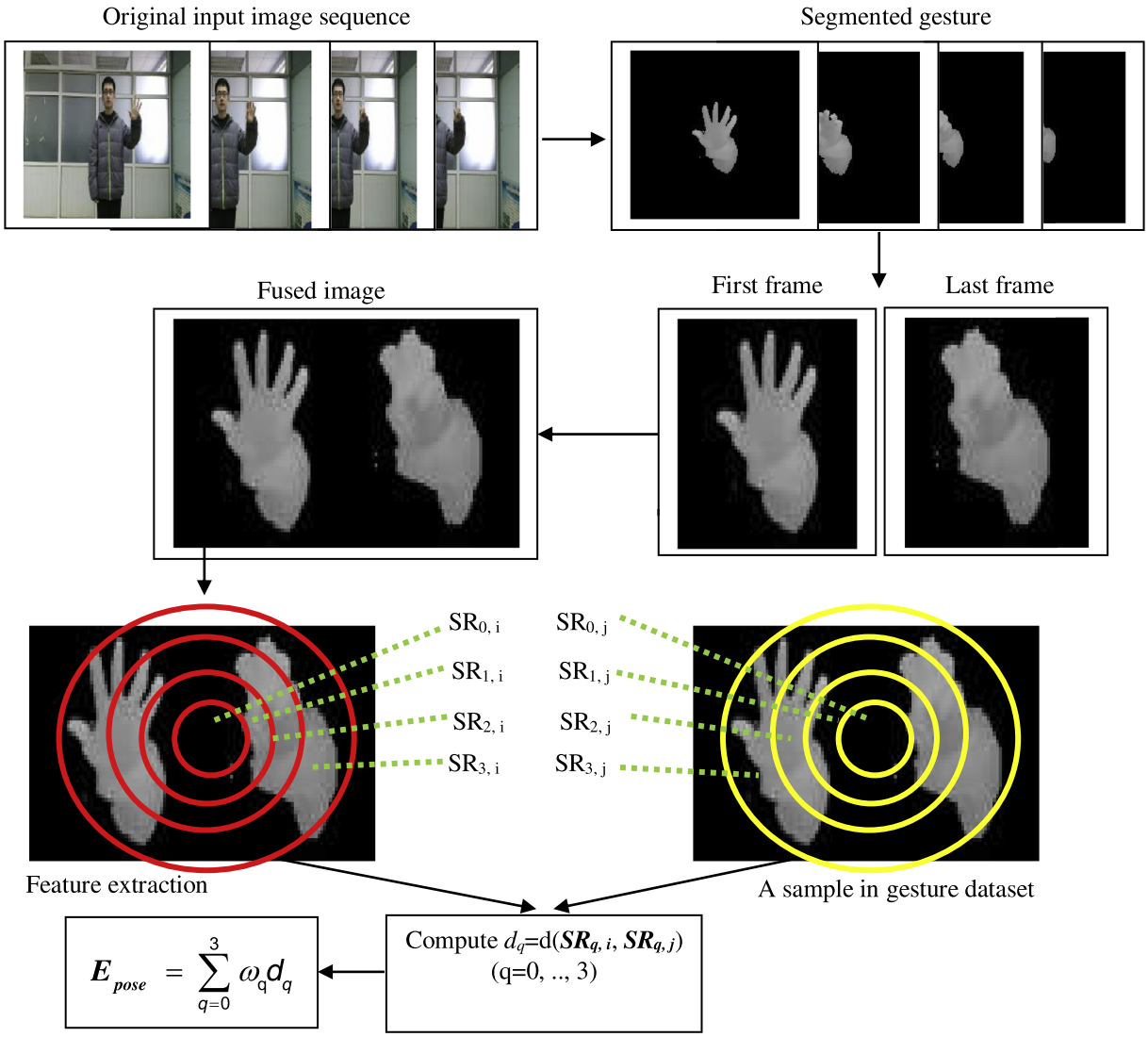


Fig. 1. Basic idea to dynamic gesture recognition.

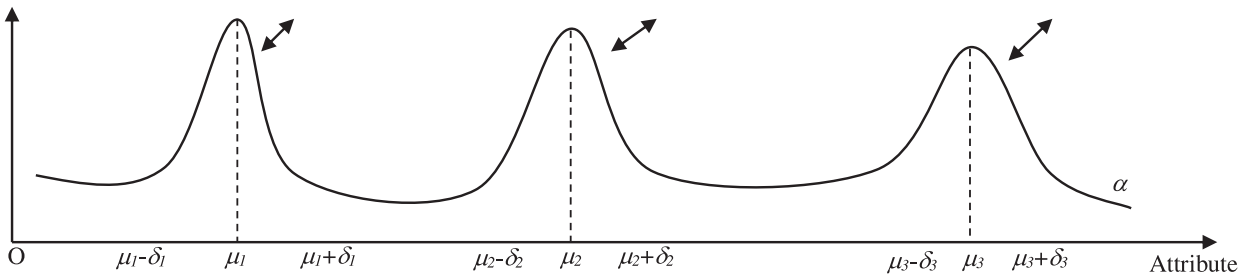


Fig. 2. The idea of ACFM.

6.2. Experimental results and analysis

6.2.1. Comparison of gesture recognition

We selected the 10 commonly used gestures in the smart teaching interface (STI) listed in Table 2. Owing to the limited length of this paper, only the FMs of the four gestures $\mathbf{G_1}$–$\mathbf{G_5}$ were determined, tested, and applied to the STI and other interesting application systems. The experimental results of $\mathbf{G_2}$ and $\mathbf{G_3}$ are provided in the appendix of the paper.

Table 2
The 10 commonly used gestures in our STS, and the G_1- G_5 were used to demonstrate the way one gesture is mapped to multiple semantics.

Gesture name	Descriptions
g_1 (G_1)	folding the fingers of an open hand into a fist
g_2(G_2)	circle drawing
g_3(G_3)	reciprocate a pointing finger
g_4(G_4)	fist gesture
g_5(G_5)	moving gesture with five fingers stretching out
g_6	grasping with the thumb and forefinger
g_7	scissor gesture
g_8	grasping with the thumb, forefinger and middle finger
g_9	stretching out the thumb and forefinger of fist gesture
g_{10}	stretching out the thumb, forefinger and middle finger of fist gesture

Table 3
The *RR* of FR is compared with that of the another two related algorithms.

Algorithm	Gesture									
	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}
FR	100%	97%	97%	99%	96%	99%	97%	97%	98%	99%
DDF [10]	88%	90%	85%	94%	80%	87%	91%	87%	83%	87%
HCDF-H [41]	92%	93%	91%	92%	90%	91%	92%	92%	90%	93%

A total of 50 college students aged 21–23 years were invited to participate in the experiment to compare the recognition rates for the three algorithms, namely, FR, DDF [10], and HCDF-H [41]. For each subject, each algorithm was performed 30 times for each gesture. Then, the recognition rate (*RR*) of a gesture is computed with the formula:

$$RR_{a,g} = \frac{n_t}{SC} \tag{18}$$

Here, S refers to the number of subjects, C refers to the times each subject does for the gesture g, and n_t refers to the number in which the gesture is successfully recognized with algorithm a. The recognition rates of FR, DDF [10], and HCDF-H [41] are listed in Table 3. In this experiment, $S = 50$, and $C = 30$.

6.2.2. Implementation of FM

Example 1. Rotate/Enlarge in STI with G_2

We obtain $SDFBM_{SG} = \{diameter\}$, $(s_1, c_1) =$ ("rotate," [30, 35]) and $(s_2, c_2) =$ ("enlarge," [40, 50]) through behavioral mode analysis (See "Task 1″ in Appendix). A participant aged 23 years with experience in operating intelligent TVs can easily rotate or enlarge the cube using the same gesture called "the circle drawing gesture" in the same interaction situation, although she was not trained in advance (Figs. 3(a)–(b)). However, two gestures are requested for the operator to perform the two functions with GC.

If the results of a gesture command execution are in accordance with the expectation or the actual intention of the operator, then we call an operation successful. If the operator operates the application system out of N attempts, and n single operations can successfully complete all the functions, then the success rate α is defined as

$$\alpha = n/N \tag{19}$$

We further invited 10 participants aged 21–25 years with experience in operating intelligent TVs to join the experiment. Each participant repeated the experiment 50 times, and the success rates are depicted in Fig. 3(c).

Example 2. Flip/Zoom in OVS (Onboard Vehicular System) with G_4

In a gesture-based OVS, the driver is constantly requested to shift gear, control the steering wheel, and manipulate the navigation interface while focusing on traffic information, thereby increasing the cognitive and operational loads of the driver. The time the eyes of the driver are fixed on the screen exceeds a specialized time. The proposed FM algorithm can effectively address this issue, with which the driver can move or zoom the navigation map with one gesture (Fig. 4(a)–(d)). In our experiment, the driver is a college student aged 22 years with experience in operating intelligent telephone, motor gesture G_4 = "fist gesture," SG = {"flip," "zoom" }, $SDFBM_{SG} = \{moving\ time,\ moving\ depth\}$, $(s_1, c_1) =$ ("flip," (gesture moving time > 3500 ms) OR (gesture moving depth > 30 cm), and $(s_2, c_2) =$ ("zoom," (gesture moving time < 2000 ms) OR (gesture moving depth < 20 cm). Only one gesture is used with FM, whereas two gestures are requested for the driver to manipulate the two functions with GC. We invited 10 participants aged 21–25 years with experience in operating intelligent TVs to join the experiment. Each participant repeated the experiment 50 times, and the success rates are illustrated in Fig. 4(e). In this example, the statistical distributions of s_1 and s_2 are as follows: $s_1 \sim N(3760, 35; 200^2, 5.3^2)$, and $s_2 \sim N(2112, 27; 105^2, 7.1^2)$.

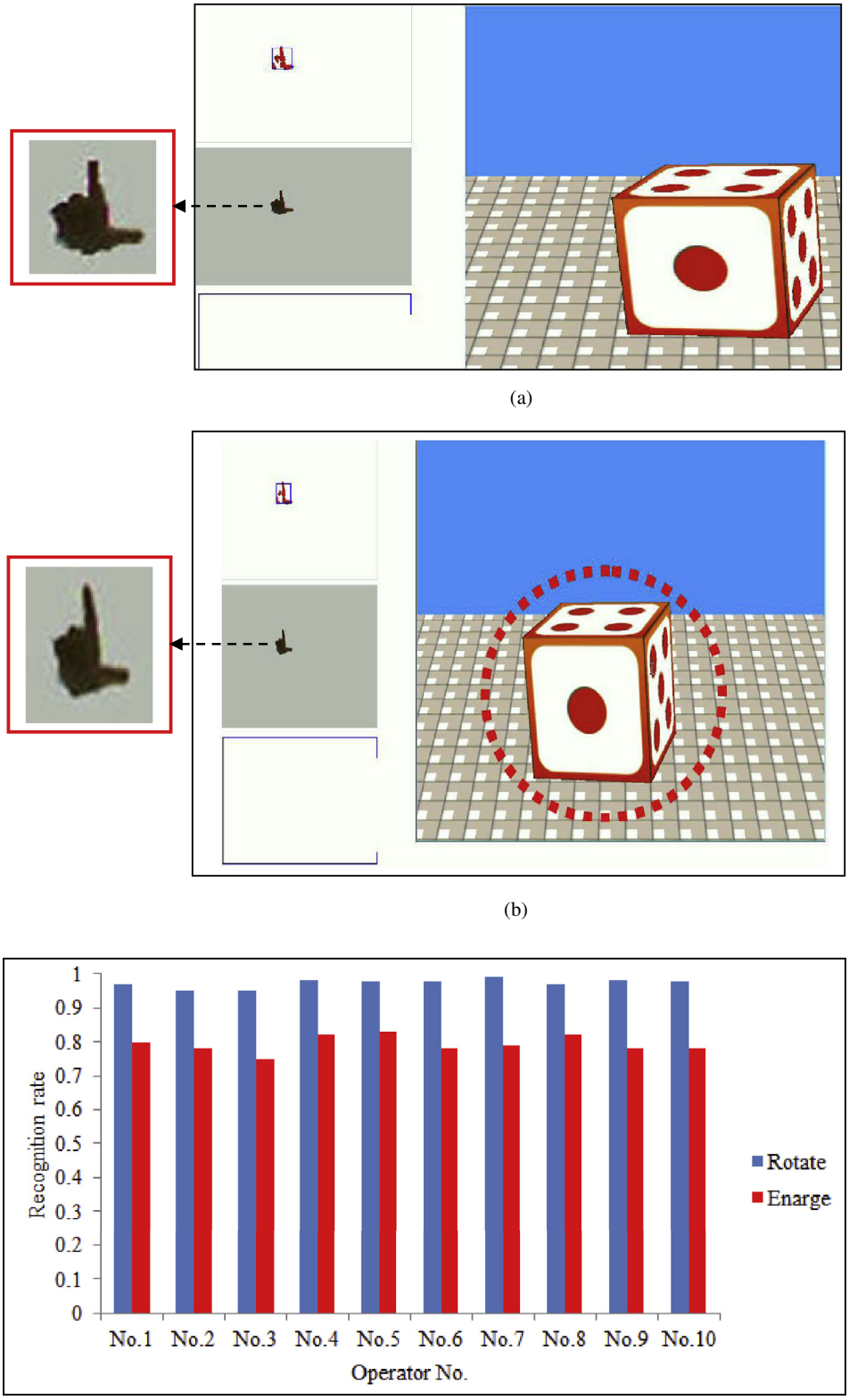


Fig. 3. Implementation of the FM between the same circle drawing gesture and the SG = (rotate, enclose) in the same interaction situation; thus, FM was verified with $\boldsymbol{SDFBM_{SG}} = \{diameter\}$. (a) to rotate the cube with the circle drawing gesture. (b) to enlarge the same object with the same gesture. (c) success rate.

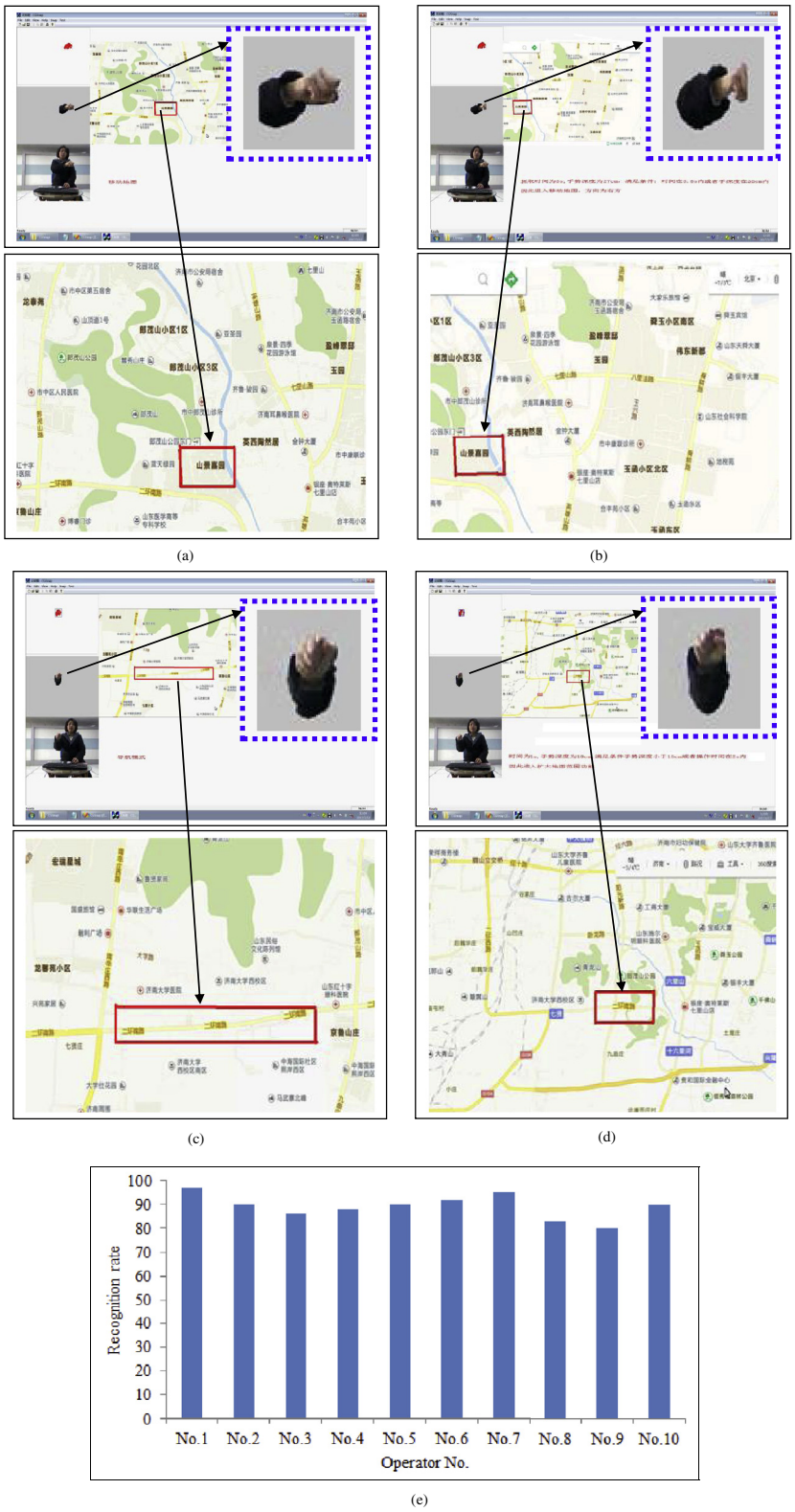


Fig. 4. FM is applied to a gesture-based on-board vehicular system. (a)-(b): at the condition of (gesture moving time > 3500 ms) OR (gesture moving depth > 30 cm), the gesture G_4 "fist gesture" is explained as the semantic "flip the map". (c)-(d): at the condition of (gesture moving time < 2000 ms) OR (gesture moving depth < 20 cm), the gesture "fist gesture" is explained as the semantic "zoom the map". (e) the success rate out of 10 participants. In (a)-(d), the bottom images are the local enlarged ones of the corresponding top images, and the images with dotted borders are the zoomed gestures.

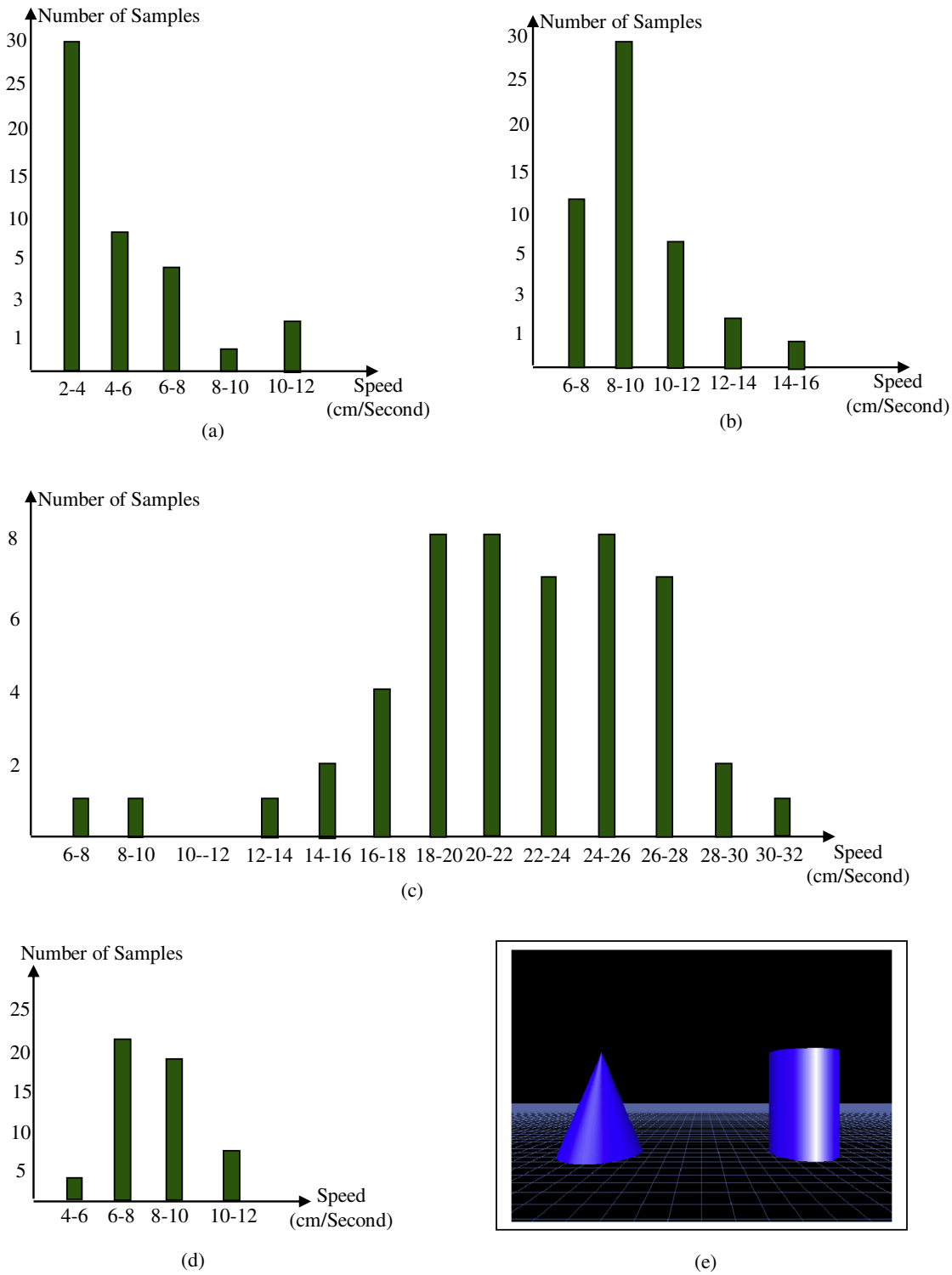


Fig. 5. Distributions of the four semantics for the gesture $\boldsymbol{G_5}$, which shows the way the gesture is mapped to four semantics in the same situational context. Here, $\boldsymbol{SG}$ = {"drag," "select," "erase," "empty-operate"}, $\boldsymbol{SDFBM_{SG}}$ = {Moving speed of $\boldsymbol{G_5}$}, (s_1, c_1) = ("drag," [2,6]), (s_2, c_2) = ("select," [8,12]), (s_3, c_3) = ("empty-operate," [21,27]), (s_4, c_4) = ("erase," [6,10]). (a) to drag (s_1). (b) to select (s_2). (c) to erase (s_3). (d) to move without any semantic (s_4). However, (s_4, c_4) and (s_2, c_2) overlapped at interval [8,10] in the same situational context; thus, the effective mapping from the gesture $\boldsymbol{G_5}$ to semantics should be $\{(s_1, c_1), (s_2, c_2), (s_3, c_3)\}$. (e) screenshot of the interactive scene.

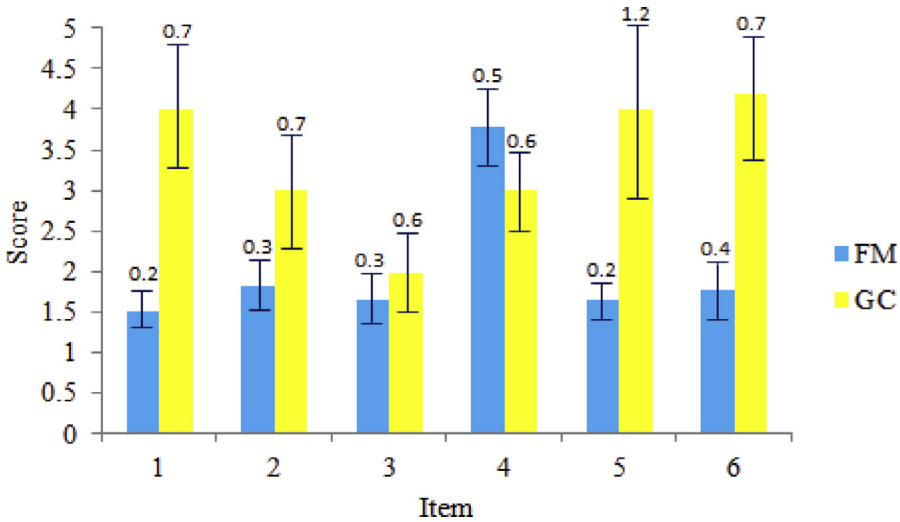


Fig. 6. Cognitive load test for FM and GC. 1–Mental demand. 2–Physical demand. 3–Temporal demand. 4–Performance. 5–Effort. 6–Frustration.

Example 3. Dragging/Selecting/Erasing/Empty-operating in STI with $\boldsymbol{G_5}$

We invited 50 subjects aged 23–26 years with experience in operating intelligent TVs to investigate the different semantics of the gesture $\boldsymbol{G_5}$. The scene has two 3D geometries as illustrated in Fig. 5e, and the subjects were requested to operate the objects with a Kinect as a gesture input device using the same gesture $\boldsymbol{G_5}$. The procedure was requested as follows. First, the subjects were instructed to drag the geometries into the center of the scene (called semantic s_1), then an object was selected (called semantic s_2), and all objects in the scene were wiped off (called semantic s_3). Lastly, the same gesture freely moves without any operating semantics (called semantic s_4). For each subject, the procedure was repeated 10 times. For each time, the average moving speed of gesture was determined, and the average motion speed of the 10 times was computed. We call the global average motion speed as a sample. Thus, we obtained 50 samples for each semantic. We attempted to map the four semantics to the gesture $\boldsymbol{G_5}$ in the same situational context. In this example, the statistical distribution obeys Gaussian normal distribution as $s_1 \sim N(4.5;\ 1.97^2)$, $s_2 \sim N(8.4;\ 1.5^2)$, $s_3 \sim N(21.6,\ 4.9^2)$, and $s_4 \sim N(9.1,\ 1.7^2)$, as demonstrated in Fig. 5(a)–(d). Here, the measurement unit is centimeters (cm). According to the distributions, we can design the flexible mappings as $(s_1,\ c_1)$=("drag," [2, 6]), $(s_2,\ c_2)$=("select," [8, 12]), $(s_3,\ c_3)$=("erase," [21, 28]), and $(s_4,\ c_4)$=("empty-operate," [6, 10]). Thus, an interesting phenomenon occurs. $(s_4,\ c_4)$ and $(s_2,\ c_2)$ overlapped at interval [8,10]. Therefore, the gesture $\boldsymbol{G_5}$ could not express the two semantics s_2 and s_4 in the same situational context. For s_4, Formula (17) cannot be satisfied. Thus, the effective mapping of the gesture $\boldsymbol{G_5}$ should be $\{(s_1,\ c_1),\ (s_2,\ c_2),\ (s_3,\ c_3)\}$.

6.2.3. Implementation of the ACFM

In Section 3.1, we aim to use the gesture $\boldsymbol{G_5}$ to test the proposed ACFM algorithm. The attribute of $\boldsymbol{G_5}$ is "moving speed," the semantic set of $\boldsymbol{G_5}$ is set as $\boldsymbol{S}$={"drag," "select," "erase"}, and each semantic in $\boldsymbol{S}$ has 50 samples. The 150 samples were divided into four categories c_i (i = 1–3) through the popular K-means clustering analysis, and $s_1 \sim N(4.5;\ 1.97^2)$, $s_2 \sim N(8.4;\ 1.5^2)$, and $s_3 \sim N(21.6,\ 4.9^2)$ were obtained by statistical analysis. Consequently, the FM from one gesture to multiple semantics is effectively generated through the ACFM algorithm.

6.2.4. User study

A large-scale comparative experiment was conducted in this study by reducing the cognitive load of the operator as the primary goal. First, 100 college juniors aged 21–24 years with experience in operating an intelligent telephone were invited to participate in the experiment. Each participant was required to experience our gesture-based SCI system, which is composed of a computer, large screen, projector, Kinect, and FM-based algorithm. Objects with 3D geometries on a large screen are manipulated with free gestures at the distance of 2–2.5 m from Kinect, which inputs and senses the gestures of the teachers. The teachers can select, scale, rotate, and translate an object with their free hand gestures, and the proposed FM algorithm is implemented in the computer. The participants are requested to implement the six functions to a cube on the large screen using bare hands before a Kinect device with FM and GC algorithms independently; these participants repeated the operation five times using each algorithm. The six functions are "to fetch," "to shrink," "to rotate," "to enclose," "to flip," and "to highlight." Each subject was requested to operate the object with the six gestures with each corresponding to a function. For FM, each subject was asked to operate the object with the three gestures with each corresponding to two functions. The gestures and SGs are displayed in Table 2 in Section 6.2. Each subject scored the operation using the NASA Task Load Index (NASA-TLX) [22] evaluation indicators on a 5-point scale. The NASA-TLX was used to quantitatively analyze and evaluate the load on operators in terms of mental demand, physical demand, time required, effort, operating

performance, degree of frustration, and other aspects; the evaluations of the corresponding indicators were averaged; the average scores of the above indicators were 1.5, 1.8, 1.6, 1.6, 3.2, and 1.7 (Fig. 6), respectively; and their mean square errors were 0.2, 0.3, 0.3, 0.5, 0.2, and 0.4, correspondingly. Fig. 6 exhibits that the FM-based interactive interface significantly reduced the cognitive and operating loads of the operator and provided advantages over the GC.

7. Conclusions and prospects

The same gesture frequently has different SDFBM features under the same situational context of an interaction; based on this interaction, a mapping between a single gesture and multiple semantics was implemented to reveal the cognitive mechanism of the FM between the gestures and interaction semantics of the operator. A method was presented to establish SG based on the behavioral models of the operator to express the different semantics with one gesture in the same interaction situation. The proposed FM significantly reduces the memory and operation workloads of the operator over the GC. The FM is further extended to the ACFM algorithm. The proposed algorithms were tested with five frequently used gestures and were applied to the STI and OVSs for complex integrated test and evaluations. Future works will focus on using random algorithms for further intelligent classification based on Gaussian normal distribution instead of Formula (17).

Acknowledgments

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Appendix. more examples for FM

A total of 50 junior college students aged 21–23 years, who had smartphone operation experience, were invited to join the experiments and perform two tasks. The test subjects wore a hand position tracking device and 5DT digital gloves, from which a computer could receive the data regarding gesture joint motions and angles through the interface.

1. Task 1: enclosing/rotating an object in STI with G_2

In Task 1, the current operation object is a 3D geometry in our STI, to which the operations of “enclose” and “rotate” can be applied. Both operations can be achieved with the gesture of “circle drawing” ($\mathbf{G_2}$) based on the interaction experience that the operators have obtained from operating smartphones. All participants joined in the “enclose” and “rotate” experiments. The 3D scene on the display had a 3D sphere, and the users were requested to use the most natural means of expressing different semantics: “enclose” and “rotate.” Each person first performed 10 times of “rotate” operations to obtain data. The 10 datasets were plotted on the same coordinate axis in MATLAB to calculate the smallest bounding box. The bounding box was further fitted into a circle through a regression method, thereby obtaining the diameter of the fitted circle of the test subject. Similarly, 10 sets of “enclose” operations were performed, and the diameter of the fitted circle of the test subject was obtained using regression.

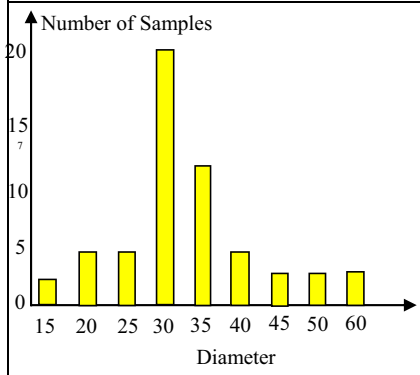
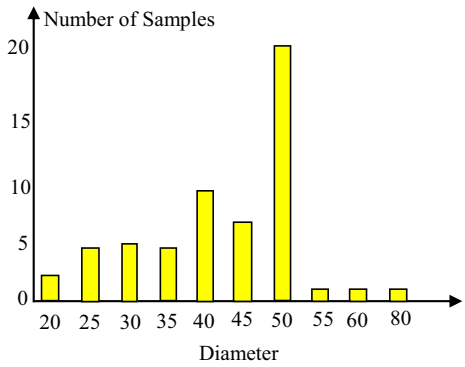
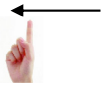
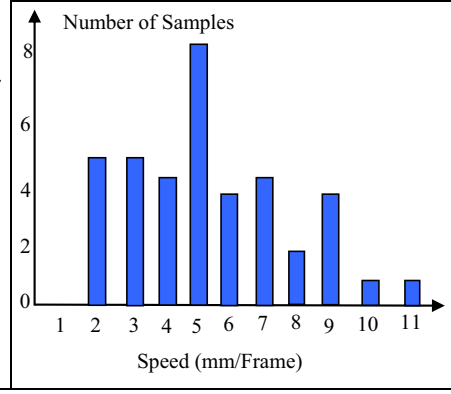
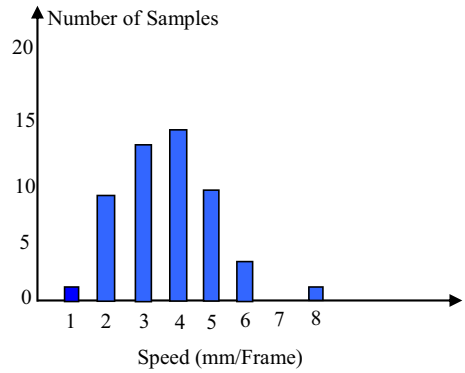
The results of Task 1 indicated that 31 of the 50 fitted circle samples, which represent 62% of all fitted circle samples, had the semantic of “rotate” and a diameter within [30,35]. A total of 34 of the 50 fitted circle samples, which consist of 78% of all fitted circle samples, had the semantic of “enclose” and a diameter within [40,50]. Thus, the two semantics for gesture $\mathbf{G_2}$ in the same situational context could be statistically distinguished using the size of the circle drawn by the gesture of the user. If the diameter of the circle was within [30,35], then the semantic of gesture $\mathbf{G_2}$ was “rotate the current object,” and if the diameter of the circle was within [40,50], then the semantic of gesture $\mathbf{G_2}$ was “enclose the current object” (Table 4a). Thus, $\boldsymbol{SDFBM_{SG}} = \{diameter\}$, $(s_1, c_1) =$ (“rotate,” [30, 35]), and $(s_2, c_2) =$ (“enclose,” [40, 50]).

The statistical analysis of the 50 samples denotes that s_1 and s_2 obey Gaussian distribution with $s_1 \sim N(32, 8.605^2)$ and $s_2 \sim N(40.32, 7.411^2)$.

2. Task 2: flipping/highlighting in STI with G_3

In Task 2, several document icons were assumed on the current page, to which two operations could be applied. One operation was that the screen page can be flipped similar to a PowerPoint slide to switch from the current to the next page. The other operation was that icons on the current page will be sequentially highlighted while the gesture is performed. Based on the experience of the operator in using smart TVs or smartphones, the operator could use the gesture “reciprocate a pointing finger ($\boldsymbol{G_3}$)” to implement both functions. In the icon-highlighting experiment, the position-tracking device was mapped to the mouse to highlight the icon pointed at by the mouse. The difference between every two frames during hand waving was recorded in each experiment, namely, the speed (unit: mm/frame). The length of each experiment was 20 frames, and the difference between each frame in the 20 frames was recorded. Therefore, 20 speed values were obtained in each experiment. Speed values were sorted in accordance with their absolute values, and the five largest and five smallest

Table 4
SDFBM features of three gestures. statistics of the SDFBM features of each gesture in the same situational context. in G_3, "1″ in the horizontal axis represents a speed within [0, 30], "2″ represents a speed within [60, 90], and so on.

Operators' gestures	Semantics of gestures		
	Rotate the current object	Enlarge the current object	
360º rotation of a pointing finger (G_2)			(a)
	Flip the ppt	Highlight the current object	
the reciprocation of a pointing finger (G_3) 			(b)

values were removed. The 10 remaining values were averaged, and the average represented the hand-waving speed of this experiment.

The results of Task 2 indicated that 27 samples, which represent 54% of all samples, for controlling the PowerPoint presentation had speeds within 120–170. For the highlighting icon task, 36 out of 50 samples, which comprise 72% of all samples, had speeds within 30–120. For all 50 groups of experiments, the overall average speed of flipping the PowerPoint presentation using hand waving was 138 mm/frame, and the overall speed of selecting icons using hand waving was 99 mm/frame (Table 4b). Table 3 displays the experimental results for using the gestures "360° rotation of a pointing finger" and "the reciprocation of a pointing finger" to express different semantics in the same situational context. For this experiment, $\boldsymbol{SDFBM_{SG}} = \{speed\}$, $(s_1, c_1) =$ ("flip," [120, 170]), and $(s_2, c_2) =$ ("highlight," [30, 120]).

The statistical analysis of the 50 samples indicates that s_1 and s_2 obey Gaussian distribution with $s_1 \sim N(155, 2.556^2)$ and $s_2 \sim N(130, 0.996^2)$.

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